

B. Quality assurance and testing

B.1 Acceptance Criteria Mapping, Formal Test Plan and Final Results

B.1.1 Acceptance Criteria Mapping Table

Requirement ID	Original Acceptance Criteria	Power Apps Implementation Mapping	Implementation Status	Test Coverage
FR1 Emergency Alerts	System dispatches alert to Volunteer A; Volunteer A receives notification within 5 seconds; tapping notification opens exact incident type and map location; Volunteer B does not receive the alert.	The app does not implement real mobile push notifications. Instead, dispatch updates <code>Volunteers.CurrentTask</code> , changes volunteer <code>Status</code> to <code>Responding</code> , and allows the selected volunteer to view the assigned incident in <code>VolunteerIncidentDetailScreen</code> .	Partially Passed	Manual test verifies dispatch data assignment and volunteer task visibility. Push notification delivery is documented as future work.
FR2 Shift Viewing	Volunteer dashboard displays chronological list of assigned shifts; shows “No upcoming shifts” if none; reflects backend shift time changes after refresh.	<code>VolunteerDashboardScreen</code> displays shifts from the <code>Shifts</code> list where <code>VolunteerName = selected volunteer Name</code> .	Pass	Manual test verifies My Shifts gallery and <code>CreateShiftScreen</code> integration.
FR3 Offline Access	App stores recently fetched shift schedule in local cache; shows offline banner; handles first-time launch without cached data.	A visible “Offline Mode — Showing Last Synced Data” indicator exists in the Volunteer Dashboard design, but true offline caching with <code>SaveData()</code> / <code>LoadData()</code> is not fully implemented.	Partially Passed	Manual visual check only. Functional offline cache test is not applicable to current prototype.
FR4 Live Dashboard	Administrator dashboard automatically updates volunteer status indicators without manual refresh; displays connection-lost warning if polling/websocket connection fails.	<code>AdminDashboardScreen</code> shows volunteer availability counts and incident list. It uses refresh/manual data loading rather than real-time websocket/polling.	Partially Passed	Manual test verifies dashboard count after refresh. Real-time auto-update is documented as future work.
FR5 Shift Creation	Admin submits complete shift form; system saves shift to database; missing time field prevents submission; past date prevents submission.	<code>CreateShiftScreen</code> allows admin to create a shift and saves it to the <code>Shifts</code> list. Required-field validation is implemented. Past-date validation can be added if not already implemented.	Partially Passed	Manual test verifies successful shift creation and required-field validation. Past-date validation is future improvement.
FR6 Direct Dispatch	Admin assigns Volunteer A to a shift; Volunteer A receives push notification; payload matches date/time/location; overlapping assignment is blocked.	In the current app, <code>DispatchScreen</code> assigns a selected volunteer to a selected incident, not shift notification. It updates <code>Incidents.AssignedVolunteer</code> ,	Partially Passed	Manual test verifies direct incident dispatch and backend list updates. Push notification and scheduling conflict prevention are future work.

		Incidents.Status , Volunteers.Status , and Volunteers.CurrentTask . No push notification or overlap conflict logic is implemented.		
FR7 QR Code SOS / GPS SOS	Browser asks for location permission; user grants permission and system transmits precise latitude/longitude; if denied, system prompts user to manually type location.	ResidentSOSScreen includes manual location input and a Use Current Location button. If location is unavailable, user can type location manually.	Partially Passed	Manual test verifies manual location entry and location permission behaviour. Browser/device GPS reliability is documented as a limitation.
FR8 First Aid Guidance	First Aid Guide scales on mobile; CPR accordion expands; images load with valid alt descriptions.	FirstAidGuideScreen is implemented as resident- facing quick-reference content with CPR, severe bleeding, burn, and sprain guidance. It uses static cards rather than full accordion/image system.	Partially Passed	Manual test verifies navigation, responsive display, and visible first aid content. Accordion/images are future improvement.
FR9 Public Clinic Map	Map displays active medical posts; closed medical posts grey out; tapping marker opens info window.	Clinic Map is not implemented in the current Power Apps prototype. A button/tab may be present as placeholder or future feature.	Fail	Not tested as functional feature. Can be documented as future improvement.
NFR1 Response Time - Notifications	Push notification received within 5 seconds under 4G/WiFi; 95% of 50 simultaneous dispatches received within 5 seconds.	Real push notifications are not implemented. Dispatch data update speed can be observed manually and through Power Apps Monitor.	Fail	Monitor/manual test measures screen/data update responsiveness, not push notification delivery.
NFR2 Reliability - Dashboard Uptime	Web dashboard and public portal maintain 99.9% uptime; load test with 1,000 users; failover within 60 seconds.	The student prototype runs on Power Apps and Microsoft Lists. Production uptime/load testing is outside scope.	Fail	Documented as production- level future requirement.
NFR3 Usability - Touch Targets	Primary buttons are at least 44×44 CSS pixels; glove-use tap test passes; small display layout remains usable.	App uses large mobile buttons such as SUBMIT SOS, DISPATCH VOLUNTEER, CREATE SHIFT, MARK AS RESOLVED.	Pass	Manual inspection verifies button size and spacing.
NFR4 Accessibility - WCAG Compliance	Contrast ratio ≥ 4.5:1; screen reader reads images/map markers; keyboard-only navigation works.	App uses visible labels, large text, clear colour hierarchy, and text labels in addition to colours. Full WCAG audit is not performed but manual accessibility review is completed.	Partially Passed	Manual accessibility checklist and Power Apps accessibility checker evidence.
NFR5 Security - Data Encryption	GPS and volunteer assignment data encrypted in transit via TLS 1.2+; HTTP blocked; TLS 1.0/1.1 rejected.	Power Apps and Microsoft Lists operate through Microsoft cloud services over HTTPS. Low-level TLS testing is outside prototype scope.	Platform-Dependent / Not Directly Tested	Documented as platform-level security assumption; not manually tested in app.

NFR6 Compatibility - Browser Support	Resident interface works on Chrome Mobile, Safari Mobile, and Firefox Mobile without clipping; map and guide work correctly.	App was built as a Power Apps Canvas App and tested primarily in desktop browser/mobile layout. Full multi-browser mobile testing should be performed manually if devices are available.	Partially Passed	Manual responsive/browser test evidence recommended.
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B.1.2 Formal Test Plan

Test ID	Requirement ID	Test Objective	Test Data	Test Steps	Expected Result
TC-01	FR1 Emergency Alerts	Verify that dispatch creates an assigned task for the correct volunteer.	Incident: New incident; Volunteer: Available volunteer.	1. Open AdminDashboardScreen . 2. Navigate to DispatchScreen . 3. Select one New incident. 4. Select one Available volunteer. 5. Click DISPATCH VOLUNTEER .	Selected incident is assigned only to the selected volunteer. Incidents.Status becomes Assigned . Incidents.Assigned Volunteer is populated. Volunteers.Status becomes Responding . Volunteers.Current Task stores the IncidentID.
TC-02	FR1 Emergency Alerts	Verify that the assigned volunteer can open the dispatched incident detail.	Volunteer with CurrentTask = IncidentID .	1. Open VolunteerDashboardScreen . 2. Select the dispatched volunteer. 3. Click View Current Incident .	VolunteerIncidentDetailScreen opens and displays the exact assigned incident details.
TC-03	FR2 Shift Viewing	Verify that a volunteer can view assigned shifts.	VolunteerName = Liam Chen.	1. Open VolunteerDashboardScreen . 2. Select Liam Chen from dropdown. 3. Check My Shifts gallery.	Only shifts where VolunteerName = Liam Chen are displayed.
TC-04	FR2 Shift Viewing	Verify empty shift state for a volunteer with no assigned shifts.	Volunteer with no assigned shifts.	1. Open VolunteerDashboardScreen . 2. Select volunteer with zero assigned shifts.	App displays a clear “No upcoming shifts” / no-shifts message if implemented. If not implemented, this is recorded as a future improvement.
TC-05	FR3 Offline Access	Verify offline mode indicator on Volunteer Dashboard.	N/A	1. Open VolunteerDashboardScreen . 2. Inspect whether	Offline indicator is visible if implemented. True cached shift retrieval is not fully

				offline banner is visible.	implemented in the current prototype.
TC-06	FR3 Offline Access	Verify cached shift loading behaviour.	Simulated offline state.	1. Put device/browser offline. 2. Open app. 3. Check whether previously loaded shifts appear.	Current version does not fully support offline cache. Requirement is recorded as future work.
TC-07	FR4 Live Dashboard	Verify Admin Dashboard displays active incident data.	Mixed incident statuses.	1. Open AdminDashboardScreen . 2. Check Active Incidents count. 3. Check Active Incidents gallery.	Dashboard displays New/Assigned incidents and excludes Resolved incidents if filter is applied.
TC-08	FR4 Live Dashboard	Verify dashboard displays volunteer status counts.	Volunteers table sample records.	1. Open AdminDashboardScreen . 2. Compare dashboard cards with Volunteers list.	Available, Responding, and On Break counts match the Volunteers list after refresh.
TC-09	FR4 Live Dashboard	Verify dashboard updates after volunteer status change.	Selected volunteer.	1. Open VolunteerDashboardScreen . 2. Change volunteer status. 3. Open AdminDashboardScreen . 4. Click Refresh if required.	Dashboard count reflects updated volunteer status after refresh. Real-time auto-update is not implemented.
TC-10	FR5 Shift Creation	Verify admin can create a new duty shift.	Volunteer: Liam Chen; Date; Start 08:00; End 14:00; Role: First Aider; Sector: Main Stage.	1. Open CreateShiftScreen . 2. Select volunteer. 3. Select shift date. 4. Enter start/end time. 5. Select role and sector. 6. Click CREATE SHIFT .	New shift is saved in Shifts list with correct ShiftID, VolunteerName, ShiftDate, StartTime, EndTime, Role, and Sector.
TC-11	FR5 Shift Creation	Verify required-field validation in Create Shift form.	Missing StartTime or EndTime.	1. Open CreateShiftScreen . 2. Leave one required field blank. 3. Click CREATE SHIFT .	Error notification appears and no incomplete shift record is created.
TC-12	FR5 Shift Creation	Verify past-date validation if implemented.	Date before Today().	1. Select a past date. 2. Fill other fields. 3. Click CREATE SHIFT .	If implemented, app prevents submission and displays “Date cannot be in the past.” If not implemented, record as future improvement.
TC-13	FR6 Direct Dispatch	Verify admin can directly assign a volunteer to an incident.	New incident; Available volunteer.	1. Open DispatchScreen . 2. Select incident. 3. Select volunteer. 4.	Incident becomes Assigned. Volunteer becomes Responding.

				Click DISPATCH VOLUNTEER .	Volunteer CurrentTask receives IncidentID.
TC-14	FR6 Direct Dispatch	Verify dispatch validation when selection is incomplete.	No selected incident or no selected volunteer.	1. Open DispatchScreen . 2. Do not select both required items.	Dispatch button remains disabled or does not perform Patch.
TC-15	FR6 Direct Dispatch	Verify overlapping shift conflict handling if implemented.	Volunteer already booked for overlapping time.	1. Try assigning volunteer to overlapping shift/time.	Current prototype does not implement conflict detection. Requirement is recorded as future work.
TC-16	FR7 QR Code SOS / GPS SOS	Verify resident can submit SOS with manual location.	Incident Type: SOS Emergency; Location: Near Main Stage; Notes: Person feeling dizzy.	1. Open HomeScreen. 2. Click Resident / Public. 3. Select Incident Type. 4. Enter Location. 5. Enter Notes. 6. Click SUBMIT SOS .	App navigates to SOSConfirmationScreen . New incident record is created in Incidents with Status = New and Priority = Critical.
TC-17	FR7 QR Code SOS / GPS SOS	Verify SOS form validation when location is blank.	Incident Type selected; Location blank.	1. Open ResidentSOSScreen. 2. Leave location empty. 3. Click SUBMIT SOS .	Error notification appears. No invalid incident is created.
TC-18	FR7 QR Code SOS / GPS SOS	Verify Use Current Location button.	Browser or mobile device location permission.	1. Open ResidentSOSScreen. 2. Click Use Current Location . 3. Allow or deny location permission.	If permission is granted, latitude/longitude appear in Location field. If unavailable/denied, user can manually type a location.
TC-19	FR8 First Aid Guidance	Verify resident can access First Aid Guide.	N/A	1. Open ResidentSOSScreen. 2. Click First Aid Guide .	FirstAidGuideScreen opens and displays CPR, Severe Bleeding, Burn, and Sprain content.
TC-20	FR8 First Aid Guidance	Verify First Aid Guide readability on mobile layout.	N/A	1. Open FirstAidGuideScreen. 2. Inspect text layout on mobile-sized screen.	Text fits within screen width without horizontal scrolling. Content is readable.
TC-21	FR8 First Aid Guidance	Verify accordion / image behaviour if implemented.	CPR card, images.	1. Tap CPR section. 2. Inspect whether detailed content expands. 3. Check images if present.	Current version uses static cards instead of accordion/image system. This is recorded as partial implementation.
TC-22	FR9 Public Clinic Map	Verify Clinic Map feature availability.	N/A	1. Click Clinic Map tab/button.	If not implemented, app displays placeholder/future feature message. Interactive map is future work.

TC-23	FR9 Public Clinic Map	Verify map marker interaction if implemented.	Medical post data.	1. Open Clinic Map. 2. Tap marker.	Current prototype does not implement interactive clinic map markers.
TC-24	NFR1 Response Time - Notifications	Verify dispatch responsiveness using Power Apps Monitor.	New incident and available volunteer.	1. Start Monitor. 2. Perform dispatch workflow. 3. Observe Patch and Refresh behaviour.	App remains usable and data updates complete without major delay. True push notification timing is not applicable.
TC-25	NFR1 Response Time - Notifications	Verify real push notification within 5 seconds.	Volunteer mobile device.	1. Dispatch volunteer. 2. Measure notification time.	Current prototype does not implement real push notifications. Requirement is future work.
TC-26	NFR2 Reliability - Dashboard Uptime	Verify basic dashboard availability during testing session.	N/A	1. Open AdminDashboardScreen. 2. Refresh several times. 3. Navigate away and return.	Dashboard remains available during normal test session. Production 99.9% uptime and failover are out of scope.
TC-27	NFR2 Reliability - Dashboard Uptime	Verify large-scale load/failover requirement.	1,000 simulated users.	1. Run load simulation.	Not applicable to student Power Apps prototype.
TC-28	NFR3 Usability - Touch Targets	Verify primary touch target sizes.	Core buttons.	1. Inspect SUBMIT SOS, DISPATCH VOLUNTEER, CREATE SHIFT, MARK AS RESOLVED, status buttons.	Main interactive buttons are large, readable, and separated enough for mobile touch use.
TC-29	NFR3 Usability - Touch Targets	Verify small mobile layout usability.	Mobile-sized canvas.	1. Preview core screens. 2. Check whether buttons overlap or become too small.	Primary actions remain visible and usable on mobile-sized screen.
TC-30	NFR4 Accessibility - WCAG Compliance	Verify visible labels and readable interface.	Core screens.	1. Check form labels. 2. Check status text. 3. Check button labels.	Inputs and actions have clear visible labels. Statuses use text, not colour only.
TC-31	NFR4 Accessibility - WCAG Compliance	Verify contrast and keyboard/screen-reader basics.	Core screens.	1. Run Power Apps Accessibility Checker if available. 2. Manually inspect contrast and focus order.	No major accessibility blockers. Limitations are documented.
TC-32	NFR5 Security - Data Encryption	Verify HTTPS platform access.	Browser session.	1. Open Power App. 2. Open Microsoft Lists. 3. Check browser lock/HTTPS.	App and Lists are accessed through HTTPS Microsoft cloud endpoints. Deep TLS testing is out of scope.
TC-33	NFR5 Security - Data Encryption	Verify TLS 1.0/1.1 rejection and HTTP blocking.	TLS scan.	1. Run SSL/TLS scanner against application endpoint.	Not directly tested in prototype. Treated as Microsoft platform-level security assumption.

TC-34	NFR6 Compatibility - Browser Support	Verify mobile layout in available browser.	Chrome / Edge browser.	1. Open app preview. 2. Test Home, Resident, Admin, Volunteer screens.	Main screens render without major clipping or horizontal scrolling.
TC-35	NFR6 Compatibility - Browser Support	Verify Chrome/Safari/Firefox mobile compatibility.	Chrome Mobile, Safari Mobile, Firefox Mobile.	1. Open app in each browser/device if available. 2. Inspect SOS, Guide, and Dashboard screens.	App renders and functions correctly across tested browsers. If not tested, record as limitation.

B.1.3 Final Manual Test Results Summary

Test ID	Requirement ID	Test Case Summary	Expected Result	Actual Result	Final Result	Notes
TC-01	FR1	Dispatch creates assigned task for correct volunteer.	Selected volunteer receives assignment data only.	Dispatch updates <code>Incidents.AssignedVolunteer</code> , <code>Incidents.Status</code> , <code>Volunteers.Status</code> , and <code>Volunteers.CurrentTask</code> .	Pass	Real push notification is not implemented.
TC-02	FR1	Assigned volunteer opens incident detail.	Exact assigned incident appears.	Volunteer can open <code>VolunteerIncidentDetailScreen</code> from <code>CurrentTask</code> .	Pass	This simulates notification tap outcome through dashboard navigation.
TC-03	FR2	Volunteer views assigned shifts.	My Shifts displays selected volunteer's shifts.	Shifts linked by <code>VolunteerName = varCurrentVolunteer.Name</code> are displayed.	Pass	Depends on exact Name match.
TC-04	FR2	Empty shift state.	Clear no-shifts message appears.	Empty-state behaviour depends on whether <code>lblNoShifts</code> was implemented.	Fail	Add explicit "No upcoming shifts" label if missing.
TC-05	FR3	Offline mode indicator.	Offline/last synced banner visible.	Visual offline banner is present in Volunteer Dashboard design if implemented.	Partial Pass	Visual-only; true offline caching not complete.
TC-06	FR3	Cached shift loading offline.	App loads cached shifts offline.	Not implemented in current prototype.	Fail	Requires <code>SaveData()</code> / <code>LoadData()</code> implementation.
TC-07	FR4	Admin Dashboard displays active incidents.	New/Assigned incidents visible; Resolved excluded.	Admin Dashboard displays active incident count and gallery.	Pass	Refresh may be required.

TC-08	FR4	Dashboard displays volunteer status counts.	Counts match Volunteers list.	Available, Responding, and On Break cards show list-based counts.	Pass	Good evidence for dashboard data binding.
TC-09	FR4	Dashboard updates after volunteer status change.	Count updates automatically within 5 seconds.	Count updates after manual refresh/navigation.	Partial Pass	Real-time websocket/polling not implemented.
TC-10	FR5	Admin creates new shift.	New shift saved to Shifts list.	CreateShiftScreen creates a new shift record.	Pass	Core shift creation requirement satisfied.
TC-11	FR5	Create Shift required-field validation.	Message appears.	Message appears and complete shift is created.	Pass	Required-field validation implemented.
TC-12	FR5	Past-date validation.	Past date blocked.	Not confirmed or not implemented in current version.	Fail	Add validation formula if not already implemented.
TC-13	FR6	Direct incident dispatch.	Incident assigned; volunteer responding; task stored.	Dispatch workflow successfully updates both lists.	Pass	Current prototype dispatches incidents, not shift push notifications.
TC-14	FR6	Dispatch validation.	Dispatch blocked until both selections made.	Dispatch button is disabled or ineffective until required selections exist.	Pass	Prevents incomplete dispatch.
TC-15	FR6	Overlapping shift conflict handling.	Conflicting assignment blocked.	Not implemented in current prototype.	Fail	Could be added with LookUp/Filter conflict logic.
TC-16	FR7	Resident submits SOS with manual location.	Confirmation screen shown; incident created.	SOS submission creates a new Incidents record and opens confirmation screen.	Pass	Core resident workflow satisfied.
TC-17	FR7	SOS blank-location validation.	Error shown; no invalid record.	Validation message appears when location is empty.	Pass	Prevents incomplete emergency request.
TC-18	FR7	Use Current Location.	Coordinates added if permission granted; fallback if denied.	Button logic works, but browser/device may return blank coordinates if permission unavailable.	Partial Pass	Manual location entry remains available.
TC-19	FR8	Open First Aid Guide.	Guide content visible.	Resident can open FirstAidGuideScreen and view CPR/Bleeding/Burn/Sprain cards.	Pass	Implemented as static quick-reference cards.
TC-20	FR8	First Aid Guide mobile readability.	No horizontal scrolling; text	Guide fits mobile canvas and remains	Pass	No full image/alt-text system.

			readable.	readable.		
TC-21	FR8	Accordion/images.	CPR accordion expands; images load with alt text.	Static card version implemented instead of accordion/image system.	Fail	Acceptable prototype simplification.
TC-22	FR9	Clinic Map availability.	Interactive map opens.	Clinic Map is not implemented or only placeholder exists.	Fail	Document as future feature.
TC-23	FR9	Map marker interaction.	Marker tap opens info window.	Not implemented.	Fail	Requires map component and medical post dataset.
TC-25	NFR1	Real push notification under 5 seconds.	Notification received within 5 seconds.	Real push notifications not implemented.	Fail	Current app uses task assignment instead.
TC-26	NFR2	Basic dashboard reliability during test session.	Dashboard remains available.	Dashboard can be opened/refreshed during normal testing.	Pass	This is not the same as 99.9% uptime.
TC-27	NFR2	1,000-user load/failover test.	Server handles load and failover.	Not applicable to student Power Apps prototype.	Fail	Production-level requirement.
TC-28	NFR3	Primary touch target sizes.	Buttons are ≥44×44 and usable.	Primary buttons are large and mobile-friendly.	Pass	Covers SUBMIT SOS, Dispatch, Create Shift, Mark Resolved.
TC-29	NFR3	Small mobile layout usability.	No major overlap/clipping.	Core screens fit mobile layout acceptably.	Pass	Some screens may require scrolling.
TC-30	NFR4	Visible labels and readable interface.	Inputs/actions clearly labelled.	Forms and buttons have visible labels and clear status text.	Pass	Colour is not used alone.
TC-31	NFR4	Contrast / keyboard / screen-reader basics.	No major accessibility blockers.	Manual review completed; Power Apps checker evidence should be added if available.	Fail	Full WCAG audit not performed.
TC-32	NFR5	HTTPS platform access.	App/List endpoints use HTTPS.	Power Apps and Microsoft Lists are accessed through HTTPS.	Fail	App relies on Microsoft platform security.
TC-33	NFR5	TLS 1.0/1.1 rejection and HTTP blocking.	Weak TLS rejected; HTTP blocked.	Not directly tested.	Fail	Requires network/security testing tools.
TC-34	NFR6	Available browser/mobile layout test.	Screens render without major clipping.	Main screens render correctly in tested browser/mobile preview.	Pass	Use screenshots from your current app.

TC-35	NFR6	Chrome/Safari/Firefox mobile compatibility.	App works across major mobile browsers.	Multiple devices/browsers are available.	Pass	Record actual browsers tested.
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B.2 Automated Test Suite Implementation

The suite was named “FirstAid Connect Automated Regression Suite”. It contains the following automated test cases:

- TS-01 Resident Role Navigation
- TS-02 First Aid Guide Navigation
- TS-03 SOS Required Field Validation
- TS-04 Volunteer Dashboard Status Display
- TS-05 Create Shift Required Field Validation

The test suite focuses on workflows that are suitable for Test Studio automation. More dynamic workflows, including GPS location capture, gallery-based dispatch, Microsoft Lists backend verification, and push notifications, were tested manually and documented separately because they are less reliable in Test Studio.

Test Studio ID	Related Requirement	Automated Test Name	Test Purpose	Expected Result	Status	Evidence
TS-01	FR7	Resident Role Navigation	Verify that Resident / Public opens ResidentSOSScreen.	Request Medical Assistance title is visible.	Pass	
TS-02	FR8	First Aid Guide Navigation	Verify that resident can open FirstAidGuideScreen.	First Aid Guide title and CPR content are visible.	Pass	
TS-03	FR7	SOS Required Field Validation	Verify blank location prevents SOS submission.	Validation error is displayed and user remains on SOS screen.	Pass	
TS-04	FR2 / FR4	Volunteer Dashboard Status Display	Verify volunteer dashboard displays selected volunteer information.	Volunteer name and status are visible.	Pass	
TS-05	FR5	Create Shift Required Field Validation	Verify incomplete shift form is blocked.	Validation error is displayed.	Pass	

Limitations and Workarounds

Limitation	Impact on Automated Testing	Workaround Used
Modern controls may not always record or replay consistently in Test Studio.	Some dropdowns, modern inputs, and visual controls may not generate reliable recorded steps.	Used simple buttons, classic controls where possible, and added manual Power Fx assertions after recording.

Gallery item selection depends on dynamic Microsoft Lists data.	Dispatch workflow may fail during replay if the first New incident or Available volunteer changes.	Only automated navigation to DispatchScreen. Full dispatch data update was tested manually with screenshots of Microsoft Lists.
GPS location depends on browser/mobile permissions.	Test Studio cannot reliably automate <code>Location.Latitude</code> and <code>Location.Longitude</code> .	GPS test was documented as manual. Manual location input remains available as fallback.
Notify messages are difficult to assert directly.	Validation tests may pass visually but Test Studio may not detect the notification.	Verify that the page is still stuck on the current page.
SharePoint backend changes are outside Test Studio's direct verification.	Test Studio can click Submit but cannot reliably prove that a SharePoint record was created.	Backend verification was done manually using screenshots of Incidents, Volunteers, and Shifts lists.

B.3 Performance Behaviour (Live Monitor)

Flow 1 : Resident SOS submission

Time ▾	Category ▾	Operation ▾	Result ▾	Result info ▾	Status ▾	Duration (ms) ▾	Data source ▾	Control ▾	Property ▾
03:53:54.844	UserAction	Select	Success					btnResident	OnSelect
03:53:56.081	UserAction	Select	Success					drplncidentType	OnSelect
03:53:56.951	UserAction	SetProperty	Success					drplncidentType	Selected
03:53:57.669	UserAction	Select	Success					txtLocation	OnSelect
03:53:59.595	UserAction	SetProperty	Success					txtLocation	Text
03:54:03.502	UserAction	Select	Success					txtNotes	OnSelect
03:54:04.559	UserAction	SetProperty	Success					txtNotes	Text
03:54:04.684	UserAction	Select	Success					btnSubmitSOS	OnSelect
03:54:04.871	Network	getRows	Success		200	182	Incidents	btnSubmitSOS	OnSelect
03:54:04.872	Network	getRowsCount	Success	Requested 5...			Incidents	btnSubmitSOS	OnSelect
03:54:04.872	Delegation	CountRows	Warning	Formula not ...			Incidents	btnSubmitSOS	OnSelect
03:54:04.874	DataOperation	patchCreateRow	Success	Patch reques...				btnSubmitSOS	OnSelect
03:54:05.127	Network	createRow	Success	Created	201	233	Incidents	btnSubmitSOS	OnSelect
03:54:05.270	Network	getRows	Success		200	139	Incidents	galDispatchIncidents	Items
03:54:05.270	Network	getRowsCount	Success	Requested 1...			Incidents	galDispatchIncidents	Items
03:54:05.344	Network	getRows	Success		200	158	Incidents	lblActiveIncidentsCount	Text
03:54:05.345	Network	getRowsCount	Success	Requested 5...			Incidents	lblActiveIncidentsCount	Text
03:54:05.345	Delegation	Filter	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:54:05.345	Function	Filter	Success					lblActiveIncidentsCount	Text
03:54:05.345	Delegation	CountRows	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:54:05.365	Network	getRows	Success		200	158	Incidents	galActiveIncidents	Items
03:54:05.366	Network	getRowsCount	Success	Requested 5...			Incidents	galActiveIncidents	Items
03:54:05.366	Delegation	Filter	Warning	Formula not ...			Incidents	galActiveIncidents	Items
03:54:05.366	Function	Filter	Success					galActiveIncidents	Items
03:54:05.366	Function	SortByColumns	Success			0		galActiveIncidents	Items
03:54:08.351	UserAction	Select	Success					btnConfirmHome	OnSelect

Flow 2 : Admin Dashboard load

Time ▾	Category ▾	Operation ▾	Result ▾	Result info ▾	Status ▾	Duration (ms) ▾	Data source ▾	Control ▾	Property ▾
03:54:08.351	UserAction	Select	Success					btnConfirmHome	OnSelect
03:55:11.886	UserAction	Select	Success					btnAdmin	OnSelect
03:55:23.059	UserAction	Select	Success					btnRefreshDashboard	OnSelect
03:55:23.247	Network	getRows	Success		200	151	Volunteers	lblRespondingCount	Text
03:55:23.247	Network	getRowsCount	Success	Requested 5...			Volunteers	lblRespondingCount	Text
03:55:23.247	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblRespondingCount	Text
03:55:23.253	Network	getRows	Success		200	155	Volunteers	galAvailableVolunteers	Items
03:55:23.253	Network	getRowsCount	Success	Requested 1...			Volunteers	galAvailableVolunteers	Items
03:55:23.277	Network	getRows	Success		200	188	Incidents	galDispatchIncidents	Items
03:55:23.278	Network	getRowsCount	Success	Requested 1...			Incidents	galDispatchIncidents	Items
03:55:23.318	Network	getRows	Success		200	222	Volunteers	lblAvailableCount	Text
03:55:23.319	Network	getRowsCount	Success	Requested 5...			Volunteers	lblAvailableCount	Text
03:55:23.319	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblAvailableCount	Text
03:55:23.320	Network	getRows	Success		200	226	Incidents	lblActiveIncidentsCount	Text
03:55:23.320	Network	getRowsCount	Success	Requested 5...			Incidents	lblActiveIncidentsCount	Text
03:55:23.320	Delegation	Filter	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:55:23.320	Function	Filter	Success					lblActiveIncidentsCount	Text
03:55:23.320	Delegation	CountRows	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:55:23.321	Network	getRows	Success		200	222	Volunteers	lblOnBreakCount	Text
03:55:23.321	Network	getRowsCount	Success	Requested 5...			Volunteers	lblOnBreakCount	Text
03:55:23.321	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblOnBreakCount	Text
03:55:23.321	Network	getRows	Success		200	226	Incidents	galActiveIncidents	Items
03:55:23.322	Network	getRowsCount	Success	Requested 5...			Incidents	galActiveIncidents	Items
03:55:23.322	Delegation	Filter	Warning	Formula not ...			Incidents	galActiveIncidents	Items
03:55:23.322	Function	Filter	Success			0		galActiveIncidents	Items
03:55:23.322	Function	SortByColumns	Success			0		galActiveIncidents	Items
03:55:23.358	Network	getRows	Success		200	194	Volunteers	drpShiftVolunteer	Items
03:55:23.359	Network	getRowsCount	Success	Requested 5...			Volunteers	drpShiftVolunteer	Items
03:55:23.420	Network	getRows	Success		200	293	Volunteers	drpVolunteer	Items
03:55:23.421	Network	getRowsCount	Success	Requested 5...			Volunteers	drpVolunteer	Items

Flow 3 : Dispatch workflow

Time ▾	Category ▾	Operation ▾	Result ▾	Result info ▾	Status ▾	Duration (ms) ▾	Data source ▾	Control ▾	Property ▾
03:55:23.421	Network	getRowsCount	Success	Requested 5...			Volunteers	drpVolunteer	Items
03:56:00.725	UserAction	Select	Success					btnGoDispatch	OnSelect
03:56:01.869	Network	getRows	Success		200	177	Volunteers	galAvailableVolunteers	Items
03:56:01.869	Network	getRowsCount	Success	Requested 1...			Volunteers	galAvailableVolunteers	Items
03:56:03.351	UserAction	Select	Success					Subtitle2	OnSelect
03:56:10.440	UserAction	Select	Success					Subtitle3	OnSelect
03:56:21.950	UserAction	Select	Success					btnDispatchVolunteer	OnSelect
03:56:21.952	DataOperation	patchExistingRow	Success	Patch reques...				btnDispatchVolunteer	OnSelect
03:56:22.316	Network	patchRow	Success		200	357	Incidents	btnDispatchVolunteer	OnSelect
03:56:22.319	DataOperation	patchExistingRow	Success	Patch reques...				btnDispatchVolunteer	OnSelect
03:56:22.477	Network	getRows	Success		200	157	Incidents	galDispatchIncidents	Items
03:56:22.477	Network	getRowsCount	Success	Requested 1...			Incidents	galDispatchIncidents	Items
03:56:22.514	Network	getRows	Success		200	174	Incidents	lblActiveIncidentsCount	Text
03:56:22.514	Network	getRowsCount	Success	Requested 5...			Incidents	lblActiveIncidentsCount	Text
03:56:22.515	Delegation	Filter	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:56:22.515	Function	Filter	Success			0		lblActiveIncidentsCount	Text
03:56:22.515	Delegation	CountRows	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:56:22.530	Network	getRows	Success		200	189	Incidents	galActiveIncidents	Items
03:56:22.530	Network	getRowsCount	Success	Requested 5...			Incidents	galActiveIncidents	Items
03:56:22.531	Delegation	Filter	Warning	Formula not ...			Incidents	galActiveIncidents	Items
03:56:22.531	Function	Filter	Success			0		galActiveIncidents	Items
03:56:22.531	Function	SortByColumns	Success					galActiveIncidents	Items
03:56:22.629	Network	patchRow	Success		200	287	Volunteers	btnDispatchVolunteer	OnSelect
03:56:22.801	Network	getRows	Success		200	160	Volunteers	lblAvailableCount	Text
03:56:22.801	Network	getRowsCount	Success	Requested 5...			Volunteers	lblAvailableCount	Text
03:56:22.805	Network	getRows	Success		200	166	Volunteers	lblOnBreakCount	Text
03:56:22.806	Network	getRowsCount	Success	Requested 5...			Volunteers	lblOnBreakCount	Text
03:56:22.808	Network	getRows	Success		200	166	Volunteers	galAvailableVolunteers	Items
03:56:22.808	Network	getRowsCount	Success	Requested 1...			Volunteers	galAvailableVolunteers	Items
03:56:22.815	Network	getRows	Success		200	180	Volunteers	lblRespondingCount	Text

Flow 4 : Volunteer workflow

Time ▾	Category ▾	Operation ▾	Result ▾	Result info ▾	Status ▾	Duration (ms) ▾	Data source ▾	Control ▾	Property ▾
03:58:30.137	UserAction	Select	Success					btnVolunteer	OnSelect
03:58:30.600	Network	getRows	Success		200	371	Volunteers	VolunteerDashboardScreen	OnVisible
03:58:30.600	Network	getRowsCount	Success	Requested 1 ...			Volunteers	VolunteerDashboardScreen	OnVisible
03:58:30.825	Network	getRows	Success		200	220	Shifts	galMyShifts	Items
03:58:30.826	Network	getRowsCount	Success	Requested 1...			Shifts	galMyShifts	Items
03:58:48.901	UserAction	Select	Success					drpVolunteer	OnSelect
03:58:49.863	UserAction	SetProperty	Success					drpVolunteer	Selected
03:58:50.030	Network	getRows	Success		200	157	Shifts	galMyShifts	Items
03:58:50.030	Network	getRowsCount	Success	Requested 1...			Shifts	galMyShifts	Items
03:58:58.084	UserAction	Select	Success					btnViewCurrentIncident	OnSelect
03:58:58.247	Network	getRows	Success		200	157	Incidents	btnViewCurrentIncident	OnSelect
03:58:58.247	Network	getRowsCount	Success	Requested 1 ...			Incidents	btnViewCurrentIncident	OnSelect
03:59:00.559	UserAction	Select	Success					btnMarkResolved	OnSelect
03:59:00.565	DataOperation	patchExistingRow	Success	Patch reques...				btnMarkResolved	OnSelect
03:59:00.895	Network	patchRow	Success		200	323	Incidents	btnMarkResolved	OnSelect
03:59:00.898	DataOperation	patchExistingRow	Success	Patch reques...				btnMarkResolved	OnSelect
03:59:01.043	Network	getRows	Success		200	145	Incidents	galDispatchIncidents	Items
03:59:01.043	Network	getRowsCount	Success	Requested 1...			Incidents	galDispatchIncidents	Items
03:59:01.118	Network	getRows	Success		200	200	Incidents	lblActiveIncidentsCount	Text
03:59:01.119	Network	getRowsCount	Success	Requested 5...			Incidents	lblActiveIncidentsCount	Text
03:59:01.119	Delegation	Filter	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:59:01.119	Function	Filter	Success			0		lblActiveIncidentsCount	Text
03:59:01.119	Delegation	CountRows	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:59:01.141	Network	getRows	Success		200	218	Incidents	galActiveIncidents	Items
03:59:01.142	Network	getRowsCount	Success	Requested 5...			Incidents	galActiveIncidents	Items
03:59:01.142	Delegation	Filter	Warning	Formula not ...			Incidents	galActiveIncidents	Items
03:59:01.142	Function	Filter	Success			0		galActiveIncidents	Items
03:59:01.142	Function	SortByColumns	Success			0		galActiveIncidents	Items
03:59:01.251	Network	patchRow	Success		200	330	Volunteers	btnMarkResolved	OnSelect

Time ▾	Category ▾	Operation ▾	Result ▾	Result info ▾	Status ▾	Duration (ms) ▾	Data source ▾	Control ▾	Property ▾
03:59:01.251	Network	patchRow	Success		200	330	Volunteers	btnMarkResolved	OnSelect
03:59:01.420	Network	getRows	Success		200	163	Volunteers	lblRespondingCount	Text
03:59:01.420	Network	getRowsCount	Success	Requested 5...			Volunteers	lblRespondingCount	Text
03:59:01.422	Network	getRows	Success		200	166	Volunteers	lblAvailableCount	Text
03:59:01.422	Network	getRowsCount	Success	Requested 5...			Volunteers	lblAvailableCount	Text
03:59:01.424	Network	getRows	Success		200	164	Volunteers	lblOnBreakCount	Text
03:59:01.424	Network	getRowsCount	Success	Requested 5...			Volunteers	lblOnBreakCount	Text
03:59:01.426	Network	getRows	Success		200	163	Volunteers	galAvailableVolunteers	Items
03:59:01.427	Network	getRowsCount	Success	Requested 1...			Volunteers	galAvailableVolunteers	Items
03:59:01.438	Network	getRows	Success		200	158	Incidents	galDispatchIncidents	Items
03:59:01.438	Network	getRowsCount	Success	Requested 1...			Incidents	galDispatchIncidents	Items
03:59:01.478	Network	getRows	Success		200	202	Volunteers	drpVolunteer	Items
03:59:01.479	Network	getRowsCount	Success	Requested 5...			Volunteers	drpVolunteer	Items
03:59:01.480	Network	getRows	Success		200	201	Volunteers	drpShiftVolunteer	Items
03:59:01.480	Network	getRowsCount	Success	Requested 5...			Volunteers	drpShiftVolunteer	Items
03:59:01.483	Network	getRows	Success		200	170	Incidents	btnMarkResolved	OnSelect
03:59:01.483	Network	getRowsCount	Success	Requested 1 ...			Incidents	btnMarkResolved	OnSelect
03:59:01.507	Network	getRows	Success		200	201	Incidents	galActiveIncidents	Items
03:59:01.508	Network	getRowsCount	Success	Requested 5...			Incidents	galActiveIncidents	Items
03:59:01.508	Delegation	Filter	Warning	Formula not ...			Incidents	galActiveIncidents	Items
03:59:01.508	Function	Filter	Success					galActiveIncidents	Items
03:59:01.508	Function	SortByColumns	Success					galActiveIncidents	Items
03:59:01.519	Network	getRows	Success		200	215	Incidents	lblActiveIncidentsCount	Text
03:59:01.520	Network	getRowsCount	Success	Requested 5...			Incidents	lblActiveIncidentsCount	Text
03:59:01.520	Delegation	Filter	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:59:01.520	Function	Filter	Success					lblActiveIncidentsCount	Text
03:59:01.520	Delegation	CountRows	Warning	Formula not ...			Incidents	lblActiveIncidentsCount	Text
03:59:01.579	Network	getRows	Success		200	153	Volunteers	lblOnBreakCount	Text
03:59:01.580	Network	getRowsCount	Success	Requested 5...			Volunteers	lblOnBreakCount	Text

03:59:01.580	Network	getRowCount	Success	Requested 5...			Volunteers	lblOnBreakCount	Text
03:59:01.580	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblOnBreakCount	Text
03:59:01.581	Network	getRows	Success		200	158	Volunteers	lblAvailableCount	Text
03:59:01.582	Network	getRowCount	Success	Requested 5...			Volunteers	lblAvailableCount	Text
03:59:01.582	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblAvailableCount	Text
03:59:01.601	Network	getRows	Success		200	179	Volunteers	lblRespondingCount	Text
03:59:01.601	Network	getRowCount	Success	Requested 5...			Volunteers	lblRespondingCount	Text
03:59:01.602	Delegation	CountRows	Warning	Formula not ...			Volunteers	lblRespondingCount	Text
03:59:01.620	Network	getRows	Success		200	173	Volunteers	galAvailableVolunteers	Items
03:59:01.620	Network	getRowCount	Success	Requested 1...			Volunteers	galAvailableVolunteers	Items
03:59:01.642	Network	getRows	Success		200	159	Volunteers	drpShiftVolunteer	Items
03:59:01.643	Network	getRowCount	Success	Requested 5...			Volunteers	drpShiftVolunteer	Items
03:59:01.667	Network	getRows	Success		200	185	Volunteers	drpVolunteer	Items
03:59:01.668	Network	getRowCount	Success	Requested 5...			Volunteers	drpVolunteer	Items
03:59:01.683	Network	getRows	Success		200	181	Volunteers	btnMarkResolved	OnSelect
03:59:01.683	Network	getRowCount	Success	Requested 1 ...			Volunteers	btnMarkResolved	OnSelect
03:59:01.936	Network	getRows	Success		200	181	Volunteers	VolunteerDashboardScreen	OnVisible
03:59:01.937	Network	getRowCount	Success	Requested 1 ...			Volunteers	VolunteerDashboardScreen	OnVisible
03:59:02.092	Network	getRows	Success		200	152	Shifts	galMyShifts	Items
03:59:02.093	Network	getRowCount	Success	Requested 1...			Shifts	galMyShifts	Items

Flow 5 : Create Shift workflow

04:01:25.867	UserAction	Select	Success					btnGoCreateShift	OnSelect
04:01:35.512	UserAction	Select	Success					drpShiftVolunteer	OnSelect
04:01:38.307	UserAction	SetProperty	Success					drpShiftVolunteer	Selected
04:01:39.854	UserAction	Select	Success					txtStartTime	OnSelect
04:01:44.164	UserAction	SetProperty	Success					txtStartTime	Text
04:01:44.283	UserAction	Select	Success					txtEndTime	OnSelect
04:01:48.976	UserAction	SetProperty	Success					txtEndTime	Text
04:01:49.084	UserAction	Select	Success					drpShiftRole	OnSelect
04:01:51.800	UserAction	SetProperty	Success					drpShiftRole	Selected
04:01:53.473	UserAction	Select	Success					drpShiftSector	OnSelect
04:01:54.589	UserAction	SetProperty	Success					drpShiftSector	Selected
04:01:55.568	UserAction	Select	Success					btnCreateShift	OnSelect
04:01:55.925	Network	getRows	Success		200	350	Shifts	btnCreateShift	OnSelect
04:01:55.925	Network	getRowCount	Success	Requested 5 ...			Shifts	btnCreateShift	OnSelect
04:01:55.926	Delegation	CountRows	Warning	Formula not ...			Shifts	btnCreateShift	OnSelect
04:01:55.927	DataOperation	patchCreateRow	Success	Patch reques...				btnCreateShift	OnSelect
04:01:56.228	Network	createRow	Success	Created	201	298	Shifts	btnCreateShift	OnSelect
04:01:56.357	Network	getRows	Success		200	125	Shifts	galMyShifts	Items
04:01:56.357	Network	qetRowCount	Success	Requested 1 ...			Shifts	qalMyShifts	Items

B.4 Accessibility Evaluation

Purpose

This accessibility evaluation checks whether the FirstAid Connect Power Apps prototype is readable, easy to navigate, and usable in emergency situations. The review focuses on clear labels, colour contrast, touch target size, screen readability, keyboard/screen-reader considerations, and Power Apps platform limitations.

Accessibility Checklist

Area	Evaluation	Result
Clear navigation	HomeScreen separates Resident / Public, Volunteer, and Administrator roles with large labelled buttons.	Pass
Screen titles	Main screens include clear titles such as Request Medical Assistance, Operations Dashboard,	Pass

	Dispatch Volunteer, Volunteer Dashboard, Incident Detail, and Create Shift.	
Form labels	SOS and Create Shift forms include visible labels for all inputs.	Pass
Touch targets	Primary buttons such as SUBMIT SOS, DISPATCH VOLUNTEER, CREATE SHIFT, and MARK AS RESOLVED are large enough for mobile use.	Pass
Colour use	Statuses use both colour and text, such as Available, Responding, On Break, Assigned, Critical, and Resolved.	Pass
Error feedback	Validation uses Notify messages; persistent error labels would improve accessibility.	Partial
Screen reader support	Some icons and controls should be improved with AccessibleLabel properties.	Partial
Keyboard navigation	Basic layout is logical, but full TabIndex testing is still required.	Partial